Grade: Check

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Stats DC27 - Assignment 1

**The probability distribution of the amount drawn from an ATM at a convenience store is tabulated below. In this table X = the amount withdrawn by a randomly selected customer and f(x) = P[X = x]. A customer can only withdraw the amounts indicated in the first column of the table below.**

| **x** | **f(x)** | x*f(x) | (x-50)^2*f(x) |
|---|---|---|---|
| **10** | **0.10** | 1.00 | -40^2*0.10 =1600*.1=160 |
| **20** | **0.15** | 3.00 | -30^2*0.15 =900*0.15=135 |
| **40** | **0.30** | 12.00 | -10^2*0.3 =30 |
| **60** | **0.15** | 9.00 | 10^2*0.15 =15 |
| **80** | **0.25** | 20.00 | 30^2*0.25 =225 |
| **100** | **0.05** | 5.00 | 50^2*0.05 =125 |
| | =1 | = 50 | = 690 |

✓ Qa

✓ Qb

✓ Qc

**a. What is the probability that a randomly selected customer withdraws at least $60 from the ATM?**

Probability that a customer withdraws at least $60 = probability of all values greater than or equal to 60:

x Qd: One-SD probability should be 0.45.

= 0.15+0.25+0.05 = 0.45 = 45%

**b. Show by calculation that the expected value withdrawn from the ATM is $50 and that the standard deviation of the amount withdrawn from this ATM is $26.27.**

1. Expected value = sum of x * f(x). Per the calculation above, the total is 50.
2. $Var(X) = \Sigma (x - \mu)^2 \cdot f(x)$, where $\mu = 50$. Square root of 690 = 26.27

**c. The manager of this convenience score tells you that an expected value of $50 is not correct because it is impossible to withdraw $50 from the ATM. How would you answer the manager?**

The expected value refers to an average across the many withdrawals over time that are expected to be made by numerous people. It is not the actual amount any one person withdraws.

**d. What is the probability that the amount withdrawn from the ATM will fall within one standard deviation of the mean?**

= μ+1*std dev = 50+ 1*26.27 = 76.27 ; μ-1*std dev = 50- 1*26.27 = 23.73

Probability of 10, 20, 40,60,80, 100 = 0.30+0.15+0.25 = 0.70 = 70%

**e. What is the probability that the amount withdrawn from the ATM will fall within two standard deviations of the mean?**

x Qe: Two-SD probability should be 1.00.

= μ+2*std dev = 50+ 2*26.27 = 102.54 ; μ-2*std dev = 50- 2*26.27 = -2.54

Probability of 40,60,80 = 0.30+0.15+0.25 = 1 = 100%s